IIT GUWAHATI BTP

A Statistical Framework for Weighted Distribution Aggregation in Heterogeneous Federated Synthetic Data Generation

Complete theoretical framework

Yes. **We will settle the mathematical framework, objectives, assumptions, and theorem targets before executing the 12 experiments.**

The distinction throughout is:

- **Definitions** specify the objects our system constructs.
- **Foundational propositions** follow from standard mathematics; I give their derivations below.
- **Method-specific guarantees** require us to finish designing the weighting method and establish its assumptions. These remain research targets.

Our working title is:

A Statistical Framework for Weighted Distribution Aggregation in Heterogeneous Federated Synthetic Data Generation

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Contents

1 The statistical setting 3

2 Observed datasets and sampling assumptions 3

3 A mathematical description of non-IID partitioning 4

4 The target population must be fixed 4

5 Local models and the release contract 5

6 The server's information constraint 6

7 Distribution discrepancies and query workloads 6

8 Aggregation and unconditional generation 7

9 The oracle problem and the expressive limit of weighting 8

10 What our statistics-based weighting method must solve 8

11 A concrete statistical weighting family we can analyze 9

12 A deterministic oracle inequality for that family **10**

13 Where the statistical uncertainty comes from **10**

14 Stability of regularized weights **11**

15 Validation oracle and honest comparison **11**

16 Client-subset generators and composition **12**

17 Conditional feature generation **13**

18 Correct conditional routing across clients **14**

19 Income prediction and its relationship to the joint model **14**

20 Reconstruction-error decomposition **15**

21 Joint-to-conditional error transfer **15**

22 Joint-to-prediction and downstream guarantees **16**

23 Query-level guarantees for generated datasets **17**

24 Consistency among the joint, conditional, and predictive models **18**

25 Privacy of the full released system **18**

26 Communication, runtime, and compression **19**

27 The complete mathematical problem **20**

28 The quantities we need to construct, estimate, or analyze **20**

29 The theorem programme and the 12 experiments **21**

1. The statistical setting

There are K clients, indexed by

$$[K] = \{1, \dots, K\}.$$

Each client holds records with the same schema:

$$Z = (X, Y) \in \mathcal{Z} = \mathcal{X} \times \mathcal{Y}.$$

For Adult,

$$\mathcal{X} = \prod_{a=1}^{d} \mathcal{X}_a, \qquad d = 14, \qquad \mathcal{Y} = \{0, 1\}.$$

Thus, a complete record contains **14 predictors and one income label**.

Some feature domains are categorical; others are numerical. We equip $\mathcal{Z}$ with a suitable sigma-algebra so that all distributions, queries, and sampling operations below are measurable.

Client i has an unknown joint distribution

$$P_i \in \mathcal{P}(\mathcal{Z}).$$

Its feature marginal and income probability are

$$P_i^X, \qquad \eta_i(x) = P_i(Y = 1 \mid X = x).$$

Heterogeneity means that, in general,

$$P_i \neq P_j.$$

This can involve differences in any of

$$P_i^X, \qquad P_i(Y), \qquad P_i(Y \mid X).$$

Our primary setting is heterogeneous clients throughout.

2. Observed datasets and sampling assumptions

Client i has three disjoint datasets:

$$D_i^{\mathrm{tr}}, \qquad D_i^{\mathrm{val}}, \qquad D_i^{\mathrm{test}},$$

with sizes

$$n_i, \qquad v_i, \qquad t_i.$$

Write

$$D_i^{\mathrm{tr}} = \{Z_{ij}\}_{j=1}^{n_i}.$$

For population-level concentration results, our basic sampling assumption is

$$Z_{ij} \overset{\mathrm{iid}}{\sim} P_i,$$

with independent held-out samples from the same client population.

This is an assumption, not a consequence of calling a partition "non-IID."

For a fixed finite dataset partitioned without replacement, some observations are dependent. We must then either use appropriate finite-population arguments or state that the target is the finite empirical population.

The empirical distributions are

$$\widehat{P}_i^{\mathrm{tr}} = \frac{1}{n_i} \sum_{j=1}^{n_i} \delta_{Z_{ij}},$$

and similarly for validation and test.

The pooled real training dataset is

$$D_R = \biguplus_{i=1}^{K} D_i^{\mathrm{tr}}, \qquad N = \sum_{i=1}^{K} n_i.$$

Here $\biguplus$ emphasizes that the client datasets are disjoint.

3. A mathematical description of non-IID partitioning

Suppose the original population is P_0. Define client-assignment probabilities

$$a_i : \mathcal{Z} \to [0,1], \qquad \sum_{i=1}^{K} a_i(z) = 1.$$

Assign each record to exactly one client using

$$I \mid Z = z \sim \mathrm{Categorical}\left(a_1(z), \ldots, a_K(z)\right).$$

The induced population proportion of client i is

$$\alpha_i^0 = \int a_i(z)\, P_0(dz).$$

Assuming $\alpha_i^0 > 0$,

$$\boxed{P_i(dz) = \frac{a_i(z)}{\alpha_i^0} P_0(dz).}$$

Consequently,

$$\boxed{P_0 = \sum_{i=1}^{K} \alpha_i^0 P_i.}$$

This formalizes partitioning using one or two features: the assignment function depends on those coordinates, while each client retains **all columns**.

A useful heterogeneity descriptor is

$$\mathsf{H}_{\mathcal{F}} = \sum_{i=1}^{K} \alpha_i d_{\mathcal{F}}(P_i, P_R),$$

where $d_{\mathcal{F}}$ is a distribution discrepancy defined below.

The partition mechanism, boundaries, random seeds, and client sizes become part of the experimental specification.

4. The target population must be fixed

Choose target proportions

$$\alpha \in \Delta_K,$$

where

$$\Delta_K = \left\{ w \in \mathbb{R}^K : w_i \geq 0,\ \sum_i w_i = 1 \right\}.$$

Define

$$\boxed{P_R = \sum_{i=1}^{K} \alpha_i P_i.}$$

For a declared pooled-record target, we can use

$$\alpha_i = \frac{n_i}{N}.$$

For the original population in the assignment construction, the appropriate proportions are α_i^0. Observed proportions n_i/N estimate those population proportions; they need not equal them exactly.

We must therefore distinguish

$$\underbrace{\alpha}_{\text{specified population proportions}} \quad \text{from} \quad \underbrace{w}_{\text{aggregation weights}}.$$

Our method changes w, while P_R remains fixed.

If α itself must be estimated, that creates another error term. For example,

$$\mathrm{TV}\left(\sum_i \alpha_i P_i, \sum_i \widehat{\alpha}_i P_i \right) \leq \frac{1}{2} \|\alpha - \widehat{\alpha}\|_1.$$

For the initial analysis, treating target proportions as fixed simplifies the problem.

5. Local models and the release contract

Each client learns a statistical joint model

$$\boxed{\widetilde{P}_i = \mathrm{Fit}_i(D_i^{\mathrm{tr}}).}$$

Our current modelling scope uses non-neural statistical methods.

A possible factorization is

$$\widetilde{P}_i(dx, dy) = \widetilde{P}_i^X(dx)\, \mathrm{Bernoulli}\left(\widehat{f}_i(x)\right)(dy),$$

where

$$\widehat{f}_i : \mathcal{X} \to [0, 1].$$

When a density or probability mass function exists,

$$\widetilde{p}_i(x, 1) = \widetilde{p}_i^X(x) \widehat{f}_i(x),$$

$$\widetilde{p}_i(x, 0) = \widetilde{p}_i^X(x) \left[1 - \widehat{f}_i(x)\right].$$

The complete client release is

$$\boxed{R_i = (\widetilde{P}_i, M_i).}$$

Here M_i is explicitly permitted metadata.

We distinguish three possibilities:

1. Public information, such as the declared schema.

2. Statistics computed from the already released model.
3. Additional statistics computed from private records.

The third category is a new private-data release and must be included in privacy accounting.

A distribution model also needs an operational interface. Depending on the method, this may support

$$\mathrm{Sample}_i(m), \qquad \mathrm{Prob}_i(C),$$
$$\mathrm{Density}_i(z), \qquad \mathrm{ConditionalSample}_i(C).$$

These capabilities are not interchangeable: sampling access alone does not imply exact density evaluation.

6. The server's information constraint

The deployable server procedure has the form

$$\boxed{\Theta = F(R_1, \ldots, R_K; U),}$$

where U denotes server randomness.

The server's output Θ may include the aggregate model, weights, conditional generator, predictor, and generated data.

It must not depend on unreleased client rows.

The real validation and test datasets belong to a separate benchmark pathway:

$$\text{client releases} \longrightarrow \text{deployable method},$$

while

$$\text{held-out real data} \longrightarrow \text{benchmark assessment}.$$

A procedure tuned directly on real validation rows is a privileged benchmark unless that access is explicitly included in the deployment protocol.

7. Distribution discrepancies and query workloads

For a class $\mathcal{F}$ of bounded measurable functions, define the integral probability discrepancy

$$d_{\mathcal{F}}(P, Q) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_P f - \mathbb{E}_Q f \right|.$$

This is an established framework for comparing distributions; the choice of function class determines what differences can be detected.[1]

Our primary operational choice is a finite workload

$$\mathcal{Q} = \{q_1, \ldots, q_L\}, \qquad q_\ell : \mathcal{Z} \to [0, 1].$$

Then

$$\boxed{d_{\mathcal{Q}}(P, Q) = \max_{\ell \in [L]} \left| \mathbb{E}_P q_\ell - \mathbb{E}_Q q_\ell \right|.}$$

Define the workload answer vector

$$b_{\mathcal{Q}}(P) = \begin{bmatrix} \mathbb{E}_P q_1 \\ \vdots \\ \mathbb{E}_P q_L \end{bmatrix}.$$

Therefore,

$$d_{\mathcal{Q}}(P,Q) = \|b_{\mathcal{Q}}(P) - b_{\mathcal{Q}}(Q)\|_\infty.$$

For a dataset,

$$b_{\mathcal{Q}}(D) = \frac{1}{|D|}\sum_{z\in D}\begin{bmatrix} q_1(z) \\ \vdots \\ q_L(z)\end{bmatrix}.$$

A finite workload gives a **pseudometric**: two different distributions can have the same answers.

For theoretical transfer results, we also use

$$\boxed{\mathrm{TV}(P,Q) = \sup_E |P(E) - Q(E)|.}$$

For densities,

$$\mathrm{TV}(P,Q) = \frac{1}{2}\int |p-q|.$$

Since our queries lie in $[0,1]$,

$$d_{\mathcal{Q}}(P,Q) \le \mathrm{TV}(P,Q).$$

The reverse inequality generally does not hold.

Thus, good performance on a finite workload does not automatically justify every total-variation-based guarantee.

8. Aggregation and unconditional generation

Let A be an aggregation operator:

$$\boxed{\widetilde{P}_{A,w} = A(\widetilde{P}_1,\dots,\widetilde{P}_K; w).}$$

It must return a valid probability distribution.

The basic family is

$$\boxed{\widetilde{P}_w = \sum_{i=1}^K w_i \widetilde{P}_i, \qquad w \in \Delta_K.}$$

Sampling is

$$I \sim \mathrm{Categorical}(w), \qquad Z \mid I = i \sim \widetilde{P}_i.$$

The synthetic dataset is

$$D_S^{(m)} = \{\widetilde{Z}_1,\dots,\widetilde{Z}_m\}, \qquad \widetilde{Z}_j \overset{\text{iid}}{\sim} \widetilde{P}_{A,w}.$$

Its empirical distribution is

$$\widehat{P}_S^{(m)} = \frac{1}{m}\sum_{j=1}^m \delta_{\widetilde{Z}_j}.$$

The mathematical objectives are different:

$$\underbrace{d_{\mathcal{Q}}(P_R, \widetilde{P}_{A,w})}_{\text{model reconstruction}},$$

and

$$\underbrace{d_{\mathcal{Q}}(P_R, \widehat{P}_S^{(m)})}_{\text{finite synthetic-data fidelity}}.$$

More generated rows reduce the second object's sampling error. They do not automatically improve the underlying model.

9. The oracle problem and the expressive limit of weighting

For a fixed A, define population reconstruction loss

$$\mathcal{L}_A(w) = d_{\mathcal{Q}}\left(P_R, \widetilde{P}_{A,w}\right).$$

The population oracle is

$$\boxed{w^*_{A,R} \in \arg\min_{w\in\Delta_K} \mathcal{L}_A(w).}$$

This is conceptual because P_R is unknown.

For mixtures, define

$$H_{\ell i} = \mathbb{E}_{\widetilde{P}_i} q_\ell, \qquad b = b_{\mathcal{Q}}(P_R).$$

Then

$$\boxed{\mathcal{L}(w) = \|Hw - b\|_\infty.}$$

The achievable workload vectors form the convex hull of the client-model vectors:

$$\{Hw : w \in \Delta_K\} = \mathrm{conv}\{H_{\cdot 1}, \ldots, H_{\cdot K}\}.$$

Therefore,

$$\boxed{\inf_{w\in\Delta_K} \mathcal{L}(w) = \mathrm{dist}_\infty\left(b, \mathrm{conv}\{H_{\cdot i}\}_{i=1}^K\right).}$$

Interpretation: weighting cannot repair every modelling error.

If b lies outside that convex hull, no mixture weights can achieve zero workload error. Improving the weighting rule and improving the local model family are separate research questions.

10. What our statistics-based weighting method must solve

Our general proposed structure is

$$S = \mathrm{Stats}(R_1, \ldots, R_K),$$

$$\boxed{\widehat{w} = T_\beta(S) \in \Delta_K.}$$

The main desired guarantee is an oracle inequality:

$$\boxed{\mathcal{L}_A(\widehat{w}) \le \inf_{w\in\Delta_K} \mathcal{L}_A(w) + \mathcal{E}_{\mathrm{weight}}.}$$

The meaningful research question is:

> Under what information and assumptions can $\mathcal{E}_{\mathrm{weight}}$ be bounded, and how does it depend on sample sizes, privacy noise, workload size, and model approximation?

Before constructing such a rule, we need an identification result.

Proposition 1: no universal model-only oracle recovery without additional assumptions.

Consider two released models on a binary variable:

$$\widetilde{P}_1 = \mathrm{Bernoulli}(1/4), \qquad \widetilde{P}_2 = \mathrm{Bernoulli}(3/4).$$

Suppose these same releases are compatible with two underlying worlds:

$$P_R^{(a)} = \mathrm{Bernoulli}(1/4),$$

$$P_R^{(b)} = \text{Bernoulli}(3/4).$$

This is possible when no restriction relates local model error to the underlying distributions.

The oracle chooses

$$w^{*(a)} = (1, 0), \qquad w^{*(b)} = (0, 1).$$

A rule receiving identical releases cannot distinguish the worlds.

Moreover,

$$\text{TV}\left(P_R^{(a)}, P_R^{(b)}\right) = \frac{1}{2}.$$

For any common output Q, the triangle inequality gives

$$\max\left\{\text{TV}(P_R^{(a)}, Q), \text{TV}(P_R^{(b)}, Q)\right\} \geq \frac{1}{4}.$$

So universal oracle recovery is impossible in this unrestricted setting.

Consequence: our theory must use an explicit assumption class, informative permitted metadata, or both.

11. A concrete statistical weighting family we can analyze

To make the theory substantive, consider the following **candidate family**, not a claim of originality.

Suppose we can construct from the releases:

$$\widehat{H} \approx H, \qquad \widehat{b} \approx b_{\mathcal{Q}}(P_R).$$

Here:

- $\widehat{H}$ describes query answers under the released client models.
- $\widehat{b}$ estimates target query answers using permitted information.

Define

$$\Omega(w) = \frac{1}{2}\|w - \alpha\|_2^2.$$

A regularized statistical weighting rule is

$$\boxed{\widehat{w}_\lambda \in \arg\min_{w \in \Delta_K}\left\{\|\widehat{H}w - \widehat{b}\|_\infty + \lambda\Omega(w)\right\}.}$$

The first term matches released target statistics. The second discourages unnecessary departures from target population proportions.

For $\lambda = 0$, this is a linear program:

$$\begin{aligned} \min_{w,u} \quad & u \\ \text{subject to} \quad & -u\mathbf{1} \leq \widehat{H}w - \widehat{b} \leq u\mathbf{1}, \\ & w \geq 0, \\ & \mathbf{1}^\top w = 1. \end{aligned}$$

For $\lambda > 0$, the objective is strongly convex and has a unique minimizer.

However, uniqueness caused by regularization does **not** establish that the underlying population weights are identifiable.

Also, fitting distributions to noisy statistical measurements is established in Private-PGM, and federated private synthetic generation under heterogeneity is addressed by FLAIM. Our eventual contribution must distinguish itself from these ideas; this optimization template alone is insufficient as a novelty claim.[2, 3]

12. A deterministic oracle inequality for that family

Assume

$$\max_i \|\widehat{H}_{\cdot i} - H_{\cdot i}\|_\infty \le r_H,$$

and

$$\|\widehat{b} - b\|_\infty \le r_b.$$

For every $w \in \Delta_K$,

$$\|(\widehat{H} - H)w\|_\infty \le \sum_i w_i \|\widehat{H}_{\cdot i} - H_{\cdot i}\|_\infty \le r_H.$$

Therefore,

$$\boxed{\left| \|\widehat{H}w - \widehat{b}\|_\infty - \|Hw - b\|_\infty \right| \le r_H + r_b.}$$

Suppose optimization has tolerance ζ:

$$\widehat{F}_\lambda(\widehat{w}_\lambda) \le \inf_{w \in \Delta_K} \widehat{F}_\lambda(w) + \zeta.$$

Let w^* minimize the unregularized population loss. Then

$$\begin{aligned} \mathcal{L}(\widehat{w}_\lambda) &\le \widehat{\mathcal{L}}(\widehat{w}_\lambda) + r_H + r_b \\ &\le \widehat{\mathcal{L}}(w^*) + \lambda\Omega(w^*) + \zeta + r_H + r_b \\ &\le \mathcal{L}(w^*) + 2(r_H + r_b) + \lambda\Omega(w^*) + \zeta. \end{aligned}$$

Hence:

$$\boxed{\mathcal{L}(\widehat{w}_\lambda) - \inf_{w \in \Delta_K} \mathcal{L}(w) \le 2(r_H + r_b) + \lambda\Omega(w^*) + \zeta.}$$

Since $\Omega(w) \le 1$ on the simplex,

$$\boxed{\mathcal{L}(\widehat{w}_\lambda) - \mathcal{L}(w^*) \le 2r_H + 2r_b + \lambda + \zeta.}$$

This is a valid initial result under the stated assumptions.

The unresolved methodological work is constructing useful $\widehat{b}$ and tight error certificates within the release budget. The inequality does not create that information for us.

13. Where the statistical uncertainty comes from

Suppose each client additionally releases a vector

$$s_i = b_{\mathcal{Q}}(\widehat{P}_i^{\mathrm{tr}}) + \xi_i,$$

where ξ_i is release noise.

Set

$$\widehat{b} = \sum_i \alpha_i s_i.$$

For fixed, data-independent queries, Hoeffding's inequality and a union bound give, with probability at least $1 - \gamma$,

$$\max_{\ell,i} \left| b_{\mathcal{Q}}(\widehat{P}_i^{\mathrm{tr}})_\ell - b_{\mathcal{Q}}(P_i)_\ell \right| \le \sqrt{\frac{\log(2KL/\gamma)}{2n_i}}$$

with the appropriate client-specific bound on the right.

Consequently,

$$\boxed{r_b \le \sum_i \alpha_i \left[\sqrt{\frac{\log(2KL/\gamma)}{2n_i}} + \|\xi_i\|_\infty \right].}$$

This decomposition separates

$$\text{finite real-sample uncertainty} \quad + \quad \text{release noise.}$$

If $\widehat{H}$ is computed from m_i generated samples per client, the corresponding bound is

$$\boxed{r_H \le \max_i \sqrt{\frac{\log(2KL/\gamma)}{2m_i}}}$$

with probability at least $1-\gamma$, conditional on the fitted models.

These two probability statements can be combined by allocating a total failure budget between them.

Two cautions matter:

- Queries adaptively selected using the same raw samples require additional analysis.
- Statistics computed only from $\widetilde{P}_i$ cannot automatically be treated as unbiased estimates of P_i.

14. Stability of regularized weights

Let two estimated objectives F and F' use the same $\lambda > 0$, and suppose

$$\sup_{w \in \Delta_K} |F(w) - F'(w)| \le \eta.$$

Let their exact minimizers be w and w'.

Strong convexity gives

$$\frac{\lambda}{2}\|w - w'\|_2^2 \le F(w') - F(w).$$

Optimality under F' implies

$$F(w') - F(w) \le 2\eta.$$

Thus,

$$\boxed{\|w - w'\|_2 \le 2\sqrt{\frac{\eta}{\lambda}}.}$$

This shows the trade-off:

$$\text{larger } \lambda \Rightarrow \text{greater stability,}$$

but potentially more bias toward α.

This is a standard strong-convexity argument. Its role is to guide the weighting design and its ablations.

15. Validation oracle and honest comparison

Define the weighted validation reference

$$\widehat{P}_R^{\text{val}} = \sum_i \alpha_i \widehat{P}_i^{\text{val}}.$$

The empirical oracle is

$$\boxed{w^*_{A,\mathrm{val}} \in \arg\min_{w\in\Delta_K} d_{\mathcal{Q}}\left(\widehat{P}^{\mathrm{val}}_R, \widetilde{P}_{A,w}\right).}$$

For fixed bounded queries and independent validation observations,

$$r_{\mathrm{val}}(\gamma) = \sqrt{\frac{1}{2}\log\left(\frac{2L}{\gamma}\right)\sum_i \frac{\alpha_i^2}{v_i}}.$$

With probability at least $1-\gamma$,

$$d_{\mathcal{Q}}\left(\widehat{P}^{\mathrm{val}}_R, P_R\right) \le r_{\mathrm{val}}.$$

By the triangle inequality, this implies uniformly over all weights

$$|\mathcal{L}_{A,\mathrm{val}}(w) - \mathcal{L}_A(w)| \le r_{\mathrm{val}}.$$

If the validation oracle is optimized to tolerance ζ_{val}, then

$$\boxed{\mathcal{L}_A(w^*_{A,\mathrm{val}}) \le \inf_w \mathcal{L}_A(w) + 2r_{\mathrm{val}} + \zeta_{\mathrm{val}}.}$$

This quantifies how good our privileged benchmark is.

The final empirical comparison uses independent test data:

$$\Delta_{\mathrm{oracle,test}} = \mathcal{L}_{A,\mathrm{test}}(\widehat{w}) - \mathcal{L}_{A,\mathrm{test}}(w^*_{A,\mathrm{val}}).$$

It can be negative because a validation oracle is not a test oracle.

If optimization is approximate and no optimality certificate exists, we must call it a **best-found benchmark**, not an exact optimum.

16. Client-subset generators and composition

Let

$$B \subseteq [K], \qquad B \neq \varnothing,$$

be a subset of clients.

Every subset generator still generates a full record in $\mathcal{Z}$.

Define

$$\widetilde{P}_B = A_B\left(\{\widetilde{P}_i : i \in B\}; w^{(B)}\right).$$

For a collection $\mathcal{B}$ of subsets,

$$\widetilde{P}_{\mathrm{sub}} = \sum_{B\in\mathcal{B}} \rho_B \widetilde{P}_B, \qquad \rho \in \Delta_{|\mathcal{B}|}.$$

If every subset generator is a mixture,

$$\widetilde{P}_B = \sum_{i\in B} w_i^{(B)} \widetilde{P}_i,$$

then

$$\boxed{\widetilde{P}_{\mathrm{sub}} = \sum_i \omega_i \widetilde{P}_i,}$$

where

$$\boxed{\omega_i = \sum_{B\ni i} \rho_B w_i^{(B)}.}$$

This follows simply by exchanging the two finite sums.

Therefore:

- Overlapping subsets require accounting for repeated client contributions.
- Exact target-mass preservation requires $\omega = \alpha$.
- Pure subset mixtures do not enlarge the original mixture family.

Any additional advantage must come from what happens **inside** the subset procedure, such as re-estimation or regularization.

17. Conditional feature generation

A condition is a measurable feature event

$$C \subseteq \mathcal{X}.$$

For rectangular conditions,

$$C = C_1 \times \cdots \times C_d,$$

where each coordinate restriction can be an interval, a category set, or the full domain.

The real target is

$$\boxed{P_R^X(dx \mid C) = \frac{\mathbf{1}_C(x) P_R^X(dx)}{P_R^X(C)}.}$$

Let $\widetilde{P}$ denote our selected final joint model. Its implied conditional is

$$\widetilde{P}^X(dx \mid C) = \frac{\mathbf{1}_C(x) \widetilde{P}^X(dx)}{\widetilde{P}^X(C)}.$$

Both require positive conditioning probability.

Our separately trained conditional generator is

$$\boxed{Q_\phi(dx \mid C) \equiv \widetilde{P}_\phi(dx \mid C).}$$

A mathematical training objective is

$$\boxed{\phi^* \in \arg\min_{\phi \in \Phi} \mathbb{E}_{C \sim \nu_{\mathrm{tr}}} \left[D\left(\widetilde{P}^X(\cdot \mid C), Q_\phi(\cdot \mid C) \right) \right],}$$

where:

- Φ is the conditional model family.
- ν_{tr} is the training distribution over conditions.
- D is an appropriate divergence or discrepancy.

The server obtains training examples from released models.

We must distinguish

$$\text{approximation error of } \Phi,$$

$$\text{finite synthetic-training error,}$$

and

$$\text{optimization error.}$$

We also measure

$$\mathrm{Viol}_\phi(C) = Q_\phi(X \notin C \mid C).$$

Condition satisfaction alone does not establish distributional fidelity within the condition.

18. Correct conditional routing across clients

For a mixture

$$\widetilde{P}_w = \sum_i w_i \widetilde{P}_i,$$

define

$$c_i(C) = \widetilde{P}_i^X(C).$$

Then

$$\widetilde{P}_w^X(C) = \sum_i w_i c_i(C).$$

The correct client-selection probability under condition C is

$$\boxed{r_i(C) = \frac{w_i c_i(C)}{\sum_j w_j c_j(C)}.}$$

Hence,

$$\boxed{\widetilde{P}_w^X(dx \mid C) = \sum_i r_i(C) \widetilde{P}_i^X(dx \mid C).}$$

A component with $c_i(C) = 0$ contributes zero.

This gives an exact reference sampler when component conditional sampling is available:

$$I \sim \text{Categorical}(r(C)),$$

$$X \mid I = i, C \sim \widetilde{P}_i^X(\cdot \mid C).$$

Selecting clients using the original w_i after imposing a condition generally produces the wrong conditional distribution.

Our trained Q_ϕ is evaluated against this implied conditional and against the real conditional reference.

19. Income prediction and its relationship to the joint model

The true target income probability is

$$\eta_R(x) = P_R(Y = 1 \mid X = x).$$

The aggregate implies

$$\boxed{\eta_{\text{joint}}(x) = \widetilde{P}(Y = 1 \mid X = x).}$$

For a mixture,

$$\boxed{\eta_{\text{joint}}(x) = \frac{\sum_i w_i \widetilde{p}_i^X(x) \widehat{f}_i(x)}{\sum_i w_i \widetilde{p}_i^X(x)},}$$

where the denominator is positive.

Define feature-dependent responsibilities

$$r_i(x) = \frac{w_i \widetilde{p}_i^X(x)}{\sum_j w_j \widetilde{p}_j^X(x)}.$$

Then

$$\eta_{\text{joint}}(x) = \sum_i r_i(x) \widehat{f}_i(x).$$

Thus, constant mixture weights lead to feature-dependent predictive contributions through Bayes' rule.

We also train a separate model

$$\boxed{\widehat{\eta}_\psi : \mathcal{X} \to [0,1].}$$

For Brier loss,

$$\psi^* \in \arg\min_{\psi\in\Psi} \mathbb{E}_{(X,Y)\sim\widetilde{P}} \left[(\widehat{\eta}_\psi(X) - Y)^2 \right].$$

At the sample level,

$$\widehat{\psi} \in \arg\min_{\psi} \frac{1}{m} \sum_{j=1}^{m} \left[\widehat{\eta}_\psi(\widetilde{X}_j) - \widetilde{Y}_j \right]^2.$$

The joint-implied predictor and the separately trained predictor are distinct objects. Their difference measures approximation and learning effects.

20. Reconstruction-error decomposition

Write

$$e_i = \mathrm{TV}(P_i, \widetilde{P}_i).$$

For a general final aggregate $\widetilde{P}_{A,w}$, define

$$a_A(w) = \mathrm{TV}\left(\sum_i w_i \widetilde{P}_i, \widetilde{P}_{A,w} \right).$$

Then

$$\boxed{\mathrm{TV}(P_R, \widetilde{P}_{A,w}) \le \sum_i \alpha_i e_i + \frac{1}{2}\|\alpha - w\|_1 + a_A(w).}$$

Derivation. Insert the two intermediate distributions

$$\sum_i \alpha_i \widetilde{P}_i, \qquad \sum_i w_i \widetilde{P}_i,$$

and apply the triangle inequality. Convexity bounds the first difference by $\sum_i \alpha_i e_i$; the mixture-weight difference is bounded by $\frac{1}{2}\|\alpha - w\|_1$.

A sharper version uses the released-model diameter

$$\widetilde{D} = \max_{i,j} \mathrm{TV}(\widetilde{P}_i, \widetilde{P}_j).$$

Then

$$\boxed{\mathrm{TV}(P_R, \widetilde{P}_{A,w}) \le \sum_i \alpha_i e_i + \frac{\widetilde{D}}{2}\|\alpha - w\|_1 + a_A(w).}$$

This captures an intuitive fact: changing weights matters less when the client models are similar.

These are upper bounds, not exact decompositions into independently observable quantities. In particular, the unknown e_i cannot generally be measured exactly from finite real samples.

21. Joint-to-conditional error transfer

Let

$$e = \mathrm{TV}(P_R, \widetilde{P}).$$

Marginalization cannot increase total variation:

$$\mathrm{TV}(P_R^X, \widetilde{P}^X) \le e.$$

Consider conditions satisfying

$$P_R^X(C) \ge \tau > e.$$

Then

$$\widetilde{P}^X(C) \ge \tau - e > 0.$$

For any event $B \subseteq \mathcal{X}$,

$$\begin{aligned}
&\left| \frac{P_R^X(B \cap C)}{P_R^X(C)} - \frac{\widetilde{P}^X(B \cap C)}{\widetilde{P}^X(C)} \right| \\
&\quad \le \frac{|P_R^X(B \cap C) - \widetilde{P}^X(B \cap C)|}{P_R^X(C)} + \frac{|P_R^X(C) - \widetilde{P}^X(C)|}{P_R^X(C)} \\
&\quad \le \frac{2e}{\tau}.
\end{aligned}$$

Taking the supremum gives

$$\boxed{\mathrm{TV}\left(P_R^X(\cdot \mid C), \widetilde{P}^X(\cdot \mid C)\right) \le \min\left\{1, \frac{2e}{\tau}\right\}.}$$

Define conditional approximation error

$$a_\phi(C) = \mathrm{TV}\left(\widetilde{P}^X(\cdot \mid C), Q_\phi(\cdot \mid C)\right).$$

Then

$$\boxed{\mathrm{TV}\left(P_R^X(\cdot \mid C), Q_\phi(\cdot \mid C)\right) \le \min\left\{1, \frac{2e}{\tau} + a_\phi(C)\right\}.}$$

This formally explains why rare conditions are difficult.

If we use rejection sampling from $\widetilde{P}^X$, the expected number of proposals is

$$\frac{1}{\widetilde{P}^X(C)} \le \frac{1}{\tau - e}.$$

A trained conditional sampler may improve efficiency, but its approximation error must be measured.

22. Joint-to-prediction and downstream guarantees

For a loss bounded in $[0, 1]$, define

$$R_P(g) = \mathbb{E}_P[\ell(g(X), Y)].$$

Then

$$\boxed{|R_{P_R}(g) - R_{\widetilde{P}}(g)| \le e.}$$

For Brier loss, the Bayes predictor is η_R, and

$$\boxed{R_{P_R}(g) - R_{P_R}(\eta_R) = \mathbb{E}_{P_R^X}[(g - \eta_R)^2].}$$

Define synthetic-distribution predictor error

$$a_\psi = \mathbb{E}_{\widetilde{P}^X}\left[(\widehat{\eta}_\psi - \eta_{\text{joint}})^2\right].$$

Using the bounded-risk transfer inequality and the optimality of the Bayes predictors,

$$\boxed{\mathbb{E}_{P_R^X}\left[(\widehat{\eta}_\psi - \eta_R)^2\right] \le a_\psi + 2e.}$$

Derivation:

$$\begin{aligned} R_{P_R}(\widehat{\eta}_\psi) &\le R_{\widetilde{P}}(\widehat{\eta}_\psi) + e \\ &= R_{\widetilde{P}}(\eta_{\text{joint}}) + a_\psi + e \\ &\le R_{\widetilde{P}}(\eta_R) + a_\psi + e \\ &\le R_{P_R}(\eta_R) + a_\psi + 2e. \end{aligned}$$

For a synthetic-trained empirical risk minimizer $g_S \in \mathcal{H}$, assume

$$\sup_{g\in\mathcal{H}} |\widehat{R}_{D_S}(g) - R_{\widetilde{P}}(g)| \le \xi_m,$$

and optimization tolerance ζ. Then

$$\boxed{R_{P_R}(g_S) - \inf_{g\in\mathcal{H}} R_{P_R}(g) \le 2e + 2\xi_m + \zeta.}$$

This separates distribution error, finite synthetic-sample learning error, and optimization error.

For the real-trained reference

$$g_R = \text{Learn}(D_R),$$

our empirical comparison is

$$\Delta_{\text{utility}} = R_{P_R}(g_S) - R_{P_R}(g_R).$$

There is **no universal guarantee that synthetic training beats real training**. We test when it helps, when it matches, and when it hurts.

These risk bounds concern the stated bounded losses. Metrics such as AUC require their own statistical treatment.

23. Query-level guarantees for generated datasets

Conditional on the fitted model, let

$$D_S^{(m)} \sim \widetilde{P}^{\otimes m}.$$

For fixed L queries, with probability at least $1-\gamma$,

$$\boxed{d_{\mathcal{Q}}\left(\widehat{P}_S^{(m)}, \widetilde{P}\right) \le \sqrt{\frac{\log(2L/\gamma)}{2m}}.}$$

Therefore,

$$\boxed{d_{\mathcal{Q}}\left(\widehat{P}_S^{(m)}, P_R\right) \le d_{\mathcal{Q}}(\widetilde{P}, P_R) + \sqrt{\frac{\log(2L/\gamma)}{2m}}.}$$

With a real test reference,

$$\begin{aligned} d_{\mathcal{Q}}\left(\widehat{P}_S^{(m)}, \widehat{P}_R^{\text{test}}\right) \le\ & d_{\mathcal{Q}}(\widetilde{P}, P_R) \\ &+ \underbrace{d_{\mathcal{Q}}(\widehat{P}_S^{(m)}, \widetilde{P})}_{\text{synthetic sampling error}} \\ &+ \underbrace{d_{\mathcal{Q}}(\widehat{P}_R^{\text{test}}, P_R)}_{\text{real reference uncertainty}}. \end{aligned}$$

For a desired synthetic sampling error at most u, it suffices that

$$\boxed{m \geq \frac{\log(2L/\gamma)}{2u^2}.}$$

This gives a principled reason for the synthetic sample sizes we later choose.

24. Consistency among the joint, conditional, and predictive models

We have three related learned objects:

$$\widetilde{P}, \qquad Q_\phi(X \mid C), \qquad \widehat{\eta}_\psi(X).$$

Predictive consistency is

$$\mathcal{C}_{\text{pred}} = \mathbb{E}_{\widetilde{P}X}\left[(\widehat{\eta}_\psi - \eta_{\text{joint}})^2\right].$$

Conditional consistency is

$$\mathcal{C}_{\text{cond}}(C) = D\left(Q_\phi(\cdot \mid C), \widetilde{P}^X(\cdot \mid C)\right).$$

For an evaluation distribution ν_{eval} over conditions,

$$\overline{\mathcal{C}}_{\text{cond}} = \mathbb{E}_{C\sim\nu_{\text{eval}}}[\mathcal{C}_{\text{cond}}(C)].$$

These measure whether our modules agree.

They do not establish agreement with reality. Consequently, we need both

internal consistency

and

external real-data fidelity.

Similarly, a low average conditional error does not imply a low worst-case conditional error. A uniform guarantee over an infinite condition class requires additional complexity assumptions.

25. Privacy of the full released system

For the initial privacy analysis, use **replace-one-record adjacency with fixed client membership and fixed public client sizes**.

Let

$$\mathcal{M}_i(D_i) = R_i.$$

The complete release is $(\varepsilon_i, \delta_i)$-differentially private if, for every neighboring pair D_i, D_i' and measurable event E,

$$\boxed{\Pr[\mathcal{M}_i(D_i) \in E] \leq e^{\varepsilon_i} \Pr[\mathcal{M}_i(D_i') \in E] + \delta_i.}$$

Post-processing preserves DP. Sequential accesses require composition; disjoint-client releases admit parallel composition under the stated adjacency model. These are established privacy properties.[4]

For a concrete metadata example, consider

$$s_i = b_{\mathcal{Q}}(\widehat{P}_i^{\text{tr}}) + \xi_i.$$

For L fixed queries in $[0,1]$, its vector-valued statistic has sensitivity

$$\Delta_1 \leq \frac{L}{n_i}.$$

Independent Laplace coordinates with scale

$$\boxed{\sigma_i = \frac{L}{n_i \varepsilon_i^{\text{stats}}}}$$

therefore provide an $\varepsilon_i^{\text{stats}}$-DP metadata mechanism.

With probability at least $1 - \gamma$,

$$\|\xi_i\|_\infty \le \sigma_i \log(L/\gamma).$$

This can be inserted directly into the weighting bound.

If the local model itself is

$$(\varepsilon_i^{\text{model}}, \delta_i^{\text{model}})\text{-DP},$$

then a basic composed release guarantee is

$$\varepsilon_i = \varepsilon_i^{\text{model}} + \varepsilon_i^{\text{stats}},$$
$$\delta_i = \delta_i^{\text{model}}.$$

For disjoint clients under fixed membership,

$$\varepsilon_{\text{overall}} = \max_i \varepsilon_i, \qquad \delta_{\text{overall}} = \max_i \delta_i.$$

All subsequent operations using only these releases—including weighting, synthetic generation, and training the server-side conditional and predictive models—are post-processing.

Important boundaries:

- Releasing a nonprivate fitted model does not become DP by adding private metadata.
- If modifying a record can move it between clients, the fixed-membership argument must be revised.
- Real validation access and published evaluation statistics need separate treatment if those records are private.
- Record-level DP is different from protecting the participation of an entire institution.

26. Communication, runtime, and compression

Define total transmitted information as

$$\boxed{B_{\text{total}} = \sum_{i=1}^{K} \Big[\text{bits}(\widetilde{P}_i) + \text{bits}(M_i)\Big] + B_{\text{additional messages}}.}$$

A release containing L numerical summaries at s bits each adds approximately

$$KLs$$

bits, excluding protocol overhead.

Runtime should be separated into

$$T_{\text{local fit}}, \quad T_{\text{release}}, \quad T_{\text{weight}},$$
$$T_{\text{generation}}, \quad T_{\text{conditional}}, \quad T_{\text{prediction}}.$$

If a compressed aggregate $\widetilde{P}^{\text{cmp}}$ satisfies

$$\text{TV}(\widetilde{P}, \widetilde{P}^{\text{cmp}}) \le e_{\text{cmp}},$$

then

$$\boxed{\text{TV}(P_R, \widetilde{P}^{\text{cmp}}) \le \text{TV}(P_R, \widetilde{P}) + e_{\text{cmp}}.}$$

Compression is another approximation term. It is not automatically a privacy guarantee.

27. The complete mathematical problem

Collect the server decisions into

$$\Theta = \left(A, \widehat{w}, \mathcal{B}, \rho, \{w^{(B)}\}, \phi, \psi\right),$$

using only components relevant to the selected method.

The fundamental information constraint is

$$\boxed{\Theta = F(R_1, \ldots, R_K; U).}$$

The primary objective is

$$\boxed{\min_F d_{\mathcal{Q}}\left(P_R, \widetilde{P}_\Theta\right).}$$

The accompanying objectives are:

$$\boxed{\mathbb{E}_{C\sim\nu_{\text{eval}}} D\left(P_R^X(\cdot \mid C), Q_\phi(\cdot \mid C)\right),}$$

$$\boxed{R_{P_R}(\widehat{\eta}_\psi) - R_{P_R}(\eta_R),}$$

and

$$\boxed{d_{\mathcal{Q}}\left(P_R, \widehat{P}_S^{(m)}\right).}$$

These are studied subject to declared constraints such as

$$B_{\text{total}} \le B_0, \qquad T_{\text{total}} \le T_0,$$

and, for the private variant,

$$\varepsilon_{\text{overall}} \le \varepsilon_0, \qquad \delta_{\text{overall}} \le \delta_0.$$

We should not combine these into an arbitrary weighted scalar objective before deciding what trade-offs the application requires.

The scientific goal is to characterize the achievable trade-off among

$$\boxed{\begin{array}{c}\text{reconstruction , conditional fidelity , predictive utility ,}\\ \text{privacy , resources.}\end{array}}$$

28. The quantities we need to construct, estimate, or analyze

Quantity	Meaning	Status
P_i	True client population	Unknown; not exactly recoverable from finite data
P_R	Declared target population	Defined by P_i and α
$\widetilde{P}_i$	Learned local joint distributions	Fit locally
M_i	Permitted metadata	Must specify content, accuracy, and privacy cost
$H, \widehat{H}$	Client-model workload answers	Evaluate exactly or estimate by sampling
$b, \widehat{b}$	True and released estimates of target workload answers	Central information requirement
T_β	Statistical weighting procedure	Main method-design object

Quantity	Meaning	Status
$\widehat{w}$	Deployable aggregation weights	Computed from releases
$w^*_{A,R}$	Population-optimal weights in family A	Theoretical comparator
$w^*_{A,\mathrm{val}}$	Validation-oracle weights	Privileged empirical benchmark
$\widetilde{P}$	Final reconstructed joint distribution	Constructed at server
$\widetilde{P}_{\mathrm{sub}}$	Composition of client-subset generators	Construct and check effective contributions
$D_S^{(m)}$	Finite synthetic dataset	Sample from final model
$Q_\phi(X \mid C)$	Trained conditional feature generator	Fit from permitted model-derived examples
$\eta_{\mathrm{joint}}(x)$	Joint-implied income probability	Derive from final joint
$\widehat{\eta}_\psi(x)$	Separately trained income predictor	Fit and evaluate
$e, e_i, a_A, a_\phi, a_\psi$	Reconstruction and module errors	Bound theoretically; diagnose empirically
ε, δ	Privacy guarantee	Derive for the implemented release
$B_{\mathrm{total}}, T_{\mathrm{total}}$	Communication and runtime	Measure and analyze

We do not need to "find the exact true distribution." We need estimators, comparisons, and justified error statements about these objects.

29. The theorem programme and the 12 experiments

The initial mathematics above gives us several foundational results already. It does **not** finish the method-specific theory.

No.	Research package	Theoretical question or guarantee
1	Aggregation and statistical weighting	Specify T_β; establish validity, optimization properties, and an oracle inequality
2	Empirical oracle benchmark	Bound validation uncertainty and optimization tolerance; compare on independent test data
3	Client-subset generators	Prove effective-weight identities; determine whether the chosen subset operation adds anything beyond mixtures
4	Unconditional generation	Separate model error and finite sampling error; determine required synthetic sample sizes
5	Trained conditional generation	Establish support conditions and joint-to-conditional transfer, including Q_ϕ's approximation error
6	Trained income probability	Analyze approximation, estimation, bounded-risk transfer, prediction, and calibration diagnostics
7	Joint-implied prediction	Derive the predictor and responsibilities; compare it with separately trained models
8	Query-level fidelity	Give simultaneous finite-workload bounds and held-out reference uncertainty
9	Consistency and ablations	Separate internal agreement from real-data fidelity; identify which components account for improvements
10	Reconstruction-error analysis	Study local model error, weighting, aggregation, heterogeneity, and sample-size dependence
11	Error transfer	Determine when joint accuracy yields conditional or predictive accuracy—and when the chosen metric is too weak

12	Privacy and resource analysis	Prove the full-release privacy statement; characterize communication, runtime, and privacy–utility trade-offs

The method-specific theory still needs four concrete choices:

1. **Which statistics are released, and why they identify useful weighting corrections.**
2. **Which exact weighting rule we propose beyond the generic minimum-distance template.**
3. **Which assumptions make its error certificate informative.**
4. **Which aspects differ substantively from existing private statistical estimation and federated synthesis methods.**

Those choices determine the paper's actual contribution. The framework above now tells us precisely what that contribution must construct, what it must prove, and what each of the 12 experiments must test.

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